

The Approach to Equilibrium: Detailed Balance and the Master Equation

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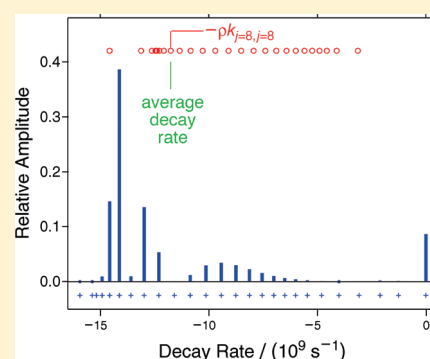
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 Supporting Information

ABSTRACT: The approach to the equilibrium (Boltzmann) distribution of populations of internal states of a molecule is governed by inelastic collisions in the gas phase and with surfaces. The set of differential equations governing the time evolution of the internal state populations is commonly called the master equation. An analytic solution to the master equation is presented and shows that the equilibrium distribution is the Boltzmann distribution. This solution is applied to the master equation involving collisions of rotational states of a diatomic molecule with a monatomic bath gas.

KEYWORDS: Upper-Division Undergraduate, Physical Chemistry, Calculator-Based Learning, Kinetic-Molecular Theory, Kinetics, Rate Law, Statistical Mechanics



At equilibrium, the populations of the internal states of a molecule are given by the Boltzmann distribution. The approach to equilibrium is governed by inelastic collisions in the gas phase and with surfaces. This energy transfer process is important in chemical kinetics, for example, the collisional deactivation of an activated molecule in a unimolecular reaction.^{1,2} The competition between relaxation and energy-dependent reaction rates will influence branching ratios and reactivity in a way that is not captured by thermal rate constants and the total concentrations of reacting species.³ An understanding of how equilibration occurs can provide insight to students into the mechanisms of complex reacting systems. The set of differential equations that govern the time evolution of populations of internal states is usually called a master equation.^{4–6}

An analytic solution to the master equation for describing collisional equilibration of a nonreacting system is presented and this method is applied to a model problem involving collisions of the rotational states of a diatomic molecule with a monatomic bath gas. In addition, MATLAB⁷ and Scilab⁸ scripts for the simulations of the temporal evolution of the rotational state populations are included. These scripts can be readily adapted to the classroom or research environment.

In a previous paper in this *Journal*, Forst⁹ introduces the concept of the master equation and outlines a similar solution of a model four-level system. The present article combines and summarizes earlier formal work in chemical kinetics and stochastic processes,^{10–12} in a way more accessible to the chemist.

MASTER EQUATION: GENERALITIES

A dilute system of diatomic molecules with M rotational states at a translational temperature T is considered. The rate of change of the number density of one state i is equal to an overall loss rate due to collisional transitions out of this state plus an infill rate due to transfer from all other states. These rates are equal to the state-to-state collisional transfer bimolecular rate constants $k_{f \rightarrow i}$ multiplied by the number density ρ of the collision partner. The rate of change of the population of state i can thus be written as

$$\frac{dn_i}{dt} = \rho[-k_{ii}n_i + \sum_{f \neq i} k_{f \rightarrow i}n_f] \quad (1)$$

or explicitly

$$\begin{aligned} \frac{dn_1}{dt} &= \rho[-k_{11}n_1 + k_{2 \rightarrow 1}n_2 + k_{3 \rightarrow 1}n_3 + \dots + k_{M \rightarrow 1}n_M] \\ \frac{dn_2}{dt} &= \rho[k_{1 \rightarrow 2}n_1 - k_{22}n_2 + k_{3 \rightarrow 2}n_3 + \dots + k_{M \rightarrow 2}n_M] \\ &\vdots \\ \frac{dn_M}{dt} &= \rho[k_{1 \rightarrow M}n_1 + k_{2 \rightarrow M}n_2 + k_{3 \rightarrow M}n_3 + \dots - k_{MM}n_M] \end{aligned} \quad (2)$$

The partial pressure of the collision partner is assumed to be so much greater than that of the target molecule, so molecule–molecule collisions can be neglected. In eq 2, the rate constants for overall loss can be distinguished by the absence of an arrow.

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This rate constant is the sum of the state-to-state rate constants out of the state:

$$k_{ii} = \sum_{f \neq i} k_{i \rightarrow f} \quad (3)$$

The coupled first-order differential equations in eq 1 or 2 can be written in matrix notation as

$$d\mathbf{n}/dt = \rho \mathbf{K} \mathbf{n} \quad (4)$$

where $\mathbf{n}$ is the column vector of the populations in states 1, ..., M and $\mathbf{K}$ is the matrix of rate constants. It is this set of coupled first-order, linear differential equations that is called a master equation. The diagonal matrix element K_{ii} is equal to the negative of the overall loss rate constant for state i , and the f th off-diagonal matrix element of $\mathbf{K}$ is the $i \rightarrow f$ rate constant. Note that the matrix $\mathbf{K}$ is real but not symmetric.

In addition, the off-diagonal matrix elements of $\mathbf{K}$ are related by the principle of detailed balance,¹³ which is the relation between the forward and backward rate constants required for the forward and backward rates to be equal when the populations are described by the Boltzmann distribution, specifically

$$g_i \exp(-E_i/k_B T) k_{i \rightarrow j} = g_j \exp(-E_j/k_B T) k_{j \rightarrow i} \quad (5)$$

In eq 5, i and j are rotational quantum numbers of two rotational states with energies E_i and E_j and degeneracies $g_i = (2i + 1)$ and $g_j = (2j + 1)$, respectively. The collisional relaxation of a system of n levels is a particular case of a system in which the transition probabilities satisfy detailed balance. This is called a reversible Markov system.¹¹

If it is assumed that the only kinetic process occurring is rotational relaxation, in particular no chemical reaction, then the total molecular number density will remain constant. To verify this, consider the rate of change of the total number density N , where $N = \sum_i n_i$. The change in the total population can be written as

$$dN/dt = \sum_i dn_i/dt = \rho \sum_i \sum_j K_{ij} n_j \quad (6)$$

Both sides of eq 6 must equal zero to conserve the total population. Each term in the sum over i is the net change in population of level i , which can have any value, depending on the current population distribution $\mathbf{n}$. Rearranging the order of the summation gives

$$dN/dt = \rho \sum_j n_j \sum_i K_{ij} \quad (7)$$

Each term in the sum over j is the sum of the elements of the j th column in the rate matrix $\mathbf{K}$ (in which the diagonal elements have a negative sign). From eq 3 it follows that each column of $\mathbf{K}$ adds to zero, and hence, the total number density N is time independent.

MASTER EQUATION: ANALYTIC SOLUTION

The master equation (eq 4) can be solved by numerical integration, for example, by using one of the ordinary differential equation (ODE) solvers in MATLAB.⁷ It is more convenient to solve these equations analytically (see also ref 4). First construct a diagonal matrix $\mathbf{\Pi}$ whose diagonal elements are the Boltzmann populations, namely,

$$\Pi_{ij} = \delta_{ij} g_i \exp(-E_i/k_B T)/Z \quad (8)$$

where Z is the partition function

$$Z = \sum_i g_i \exp(-E_i/k_B T) \quad (9)$$

The square root of this matrix, $\mathbf{\Pi}^{1/2}$, is also a diagonal matrix with elements

$$\Pi_{ij}^{1/2} = \delta_{ij} [\Pi_{ii}]^{1/2} \quad (10)$$

Because $\mathbf{\Pi}^{1/2}$ is a diagonal matrix without any zeros in the diagonal, its inverse exists and is the diagonal matrix of reciprocals. Thus, $\mathbf{\Pi}^{1/2}(\mathbf{\Pi}^{1/2})^{-1} = \mathbf{I}$, the identity matrix. Insertion of this expression for the identity matrix allows eq 4 to be rewritten as¹⁰

$$d\mathbf{n}/dt = \rho \mathbf{\Pi}^{1/2} [(\mathbf{\Pi}^{1/2})^{-1} \mathbf{K} \mathbf{\Pi}^{1/2}] (\mathbf{\Pi}^{1/2})^{-1} \mathbf{n} \quad (11)$$

Premultiplying both sides of eq 11 by $(\mathbf{\Pi}^{1/2})^{-1}$,

$$d\tilde{\mathbf{n}}/dt = \rho [(\mathbf{\Pi}^{1/2})^{-1} \mathbf{K} \mathbf{\Pi}^{1/2}] \tilde{\mathbf{n}} = \rho \tilde{\mathbf{K}} \tilde{\mathbf{n}} \quad (12)$$

where

$$\tilde{\mathbf{K}} = (\mathbf{\Pi}^{1/2})^{-1} \mathbf{K} \mathbf{\Pi}^{1/2} \quad (13)$$

The new number density vector $\tilde{\mathbf{n}}$ is related to the original set of number densities by

$$\tilde{\mathbf{n}} = (\mathbf{\Pi}^{1/2})^{-1} \mathbf{n} \quad (14)$$

The diagonal elements of the matrix $\tilde{\mathbf{K}}$ are identical to those of the original rate matrix $\mathbf{K}$. The off-diagonal elements are given by

$$\tilde{k}_{i \rightarrow j} = \left(\frac{g_j}{g_i} \right)^{1/2} \exp[-(E_j - E_i)/2k_B T] k_{i \rightarrow j}$$

From the principle of detailed balance (eq 5), one can show that $\tilde{\mathbf{K}}$ is symmetric.¹¹ Thus, $\tilde{\mathbf{K}}$ will have real eigenvalues and can be diagonalized by an orthogonal transformation $\mathbf{D}$:

$$\mathbf{D}^T \tilde{\mathbf{K}} \mathbf{D} = \mathbf{\Lambda} \quad (15)$$

where $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues Λ_i and where

$$\mathbf{D}^T \mathbf{D} = \mathbf{D} \mathbf{D}^T = \mathbf{I} \quad (16)$$

The columns of $\mathbf{D}$ are the eigenvectors of $\tilde{\mathbf{K}}$. Because $\mathbf{K}$ and $\tilde{\mathbf{K}}$ are related by a similarity transformation, $\mathbf{K}$ will have the same real eigenvalues as $\tilde{\mathbf{K}}$. However, the eigenvectors of $\mathbf{K}$ and $\tilde{\mathbf{K}}$ are not the same but are related by eq 14. In particular, the eigenvectors of $\tilde{\mathbf{K}}$ are orthogonal, whereas those of $\mathbf{K}$ are not.

Equation 12 can now be simplified to read

$$d\tilde{\mathbf{n}}/dt = \rho \tilde{\mathbf{K}} \tilde{\mathbf{n}} = \rho \mathbf{D} \mathbf{D}^T \tilde{\mathbf{K}} \mathbf{D} \mathbf{D}^T \tilde{\mathbf{n}} = \rho \mathbf{D} \mathbf{\Lambda} \mathbf{D}^T \tilde{\mathbf{n}} \quad (17)$$

After premultiplying by $\mathbf{D}^T$,

$$d\mathcal{N}/dt = \rho \mathbf{\Lambda} \mathcal{N} \quad (18)$$

where

$$\mathcal{N} = \mathbf{D}^T \tilde{\mathbf{n}} = \mathbf{D}^T (\mathbf{\Pi}^{1/2})^{-1} \mathbf{n} \quad (19)$$

Equation 18 is a diagonal set of uncoupled differential equations. Hence, each component of the new population vector $\mathcal{N}$ evolves independently and, from eq 18, relaxes exponentially, namely,

$$\mathcal{N}_i(t) = \mathcal{N}_i(0) \exp(\rho \Lambda_i t) \quad (20)$$

Each component of the vector $\mathcal{N}$ is, from eq 18, a linear

Table 1. Parameters Used in the Simulation of the Solution of the Master Equation

Parameter ^a	Value
M	26
B	8.115 cm^{-1}
α	$3.76 \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$
β	2.00
T	800 K
ρ	$9.2 \times 10^{18} \text{ molecules cm}^{-3}$

^aThese parameters are appropriate to collisions of HBr($\nu = 1$) with helium at 1 atm and 800 K. The scaling law parameters were taken from ref 19.

combination of the level populations, n_i . Because each of these components relaxes independently, they can be referred to as “normal modes” of the relaxation.

It turns out that one of the eigenvalues of the rate constant matrix is zero while all the others are negative. The second conclusion is a consequence of the Gershgorin circle theorem,¹⁴ which states that for any complex $N \times N$ matrix, all eigenvalues lie within at least one of the so-called Gershgorin disks. The i th disk is a closed circle centered at the diagonal matrix element K_{ii} with radius R_i equal to $\sum_{j \neq i} |K_{ij}|$. Because all the off-diagonal elements of $\mathbf{K}$ are positive, $R_i = \sum_{j \neq i} K_{ij}$. Moreover, each diagonal element of $\mathbf{K}$ is equal to the negative of the sum of the off-diagonal elements in column i . Thus, $R_i = |K_{ii}|$. Consequently, each Gershgorin disk is centered at $K_{ii} = -k_{ii}$ and is just large enough to touch the origin from below. Hence, all the eigenvalues of $\mathbf{K}$ are less than or equal to zero; in other words, the matrix $\mathbf{K}$ is negative semi-definite.

The proof that one eigenvalue of $\mathbf{K}$ is zero is straightforward. Because each diagonal matrix element of the $\mathbf{K}$ matrix is the negative of the sum of all the off-diagonal matrix elements in the same column (eq 3), this matrix is singular, and hence, the determinant of the matrix is zero. (This statement can readily be verified for the case of a 2- or 3-level system by evaluation of the determinant of the $\mathbf{K}$ matrix.) This implies that its secular determinant vanishes, so that one of the eigenvalues is zero. In the Supporting Information, the appendix shows that (i) the zero eigenvalue of the matrix $\tilde{\mathbf{K}}$, and hence of $\mathbf{K}$, is nondegenerate and (ii) the elements of the corresponding eigenvector of $\mathbf{K}$ are the Boltzmann populations.

The relaxation of an initial state distribution $\mathbf{n}_0$ can be expressed as follows: Without loss of generality, first assume that this initial state distribution is normalized, so that $\sum_i (n_0)_i = 1$. Then re-express the $\mathbf{n}_0$ distribution in terms of the eigenvectors of $\tilde{\mathbf{K}}$, by projecting the initial distribution onto the relaxation normal modes, namely,¹¹

$$\mathcal{N}_0 = \mathbf{D}^T \tilde{\mathbf{n}}_0 = \mathbf{D}^T (\mathbf{\Pi}^{1/2})^{-1} \mathbf{n}_0 \quad (21)$$

As time proceeds, the components of $\mathcal{N}_0$ each decay exponentially, so that the state populations can be written as

$$\mathbf{n}(t) = \mathbf{\Pi}^{1/2} \mathbf{D} \mathcal{N}_0 \exp(\rho \Lambda t) \quad (22)$$

Here, $\mathcal{N}_0 \exp(\rho \Lambda t)$ is a column vector, with elements $\mathcal{N}_i(t=0) \exp(\rho \Lambda_i t)$. At long times, all the components corresponding to nonzero values of Λ_i decay to zero. (They all decay because all the nonzero eigenvalues of the rate matrix are negative.) Only the component with zero eigenvalue survives. But, as

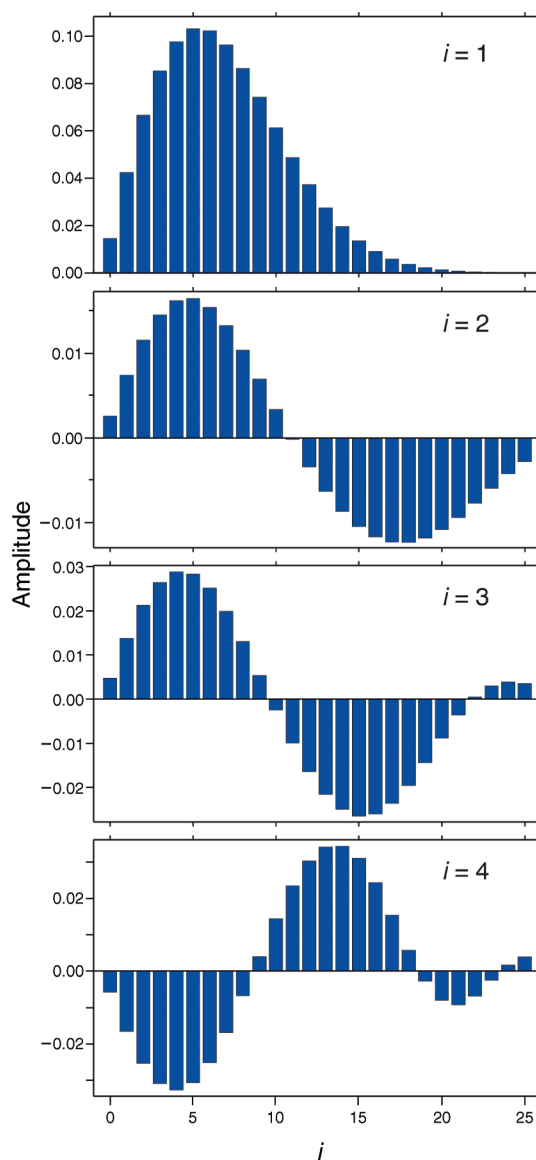


Figure 1. Lowest 4 eigenvectors of the $\mathbf{K}$ matrix. The lowest ($i = 1$) corresponds to the Boltzmann distribution. The other eigenvectors (not shown here) are the higher normal modes of the relaxation.

mentioned above (and proved in the Supporting Information), the state distribution corresponding to this eigenvalue of $\mathbf{K}$ is the Boltzmann distribution. Consequently, as anticipated, collisional relaxation leads in the long-time limit to a Boltzmann distribution.

MODEL PROBLEM: ROTATIONAL EQUILIBRATION OF A DIATOMIC MOLECULE

To illustrate the details of collisional relaxation, a program (master_equation_hbr.m, coded in MATLAB,⁷ or master_equation_hbr.sce, coded in the public-domain Scilab⁸ package), was written that solves, following the diagonalization procedure presented in the previous section, the master equation for the relaxation of M coupled rotational states, ranging from $j = 0$ to $j = M - 1$, with degeneracies $2j + 1$ and energies $E_j = B j(j + 1)$. The rate matrix is modeled by an

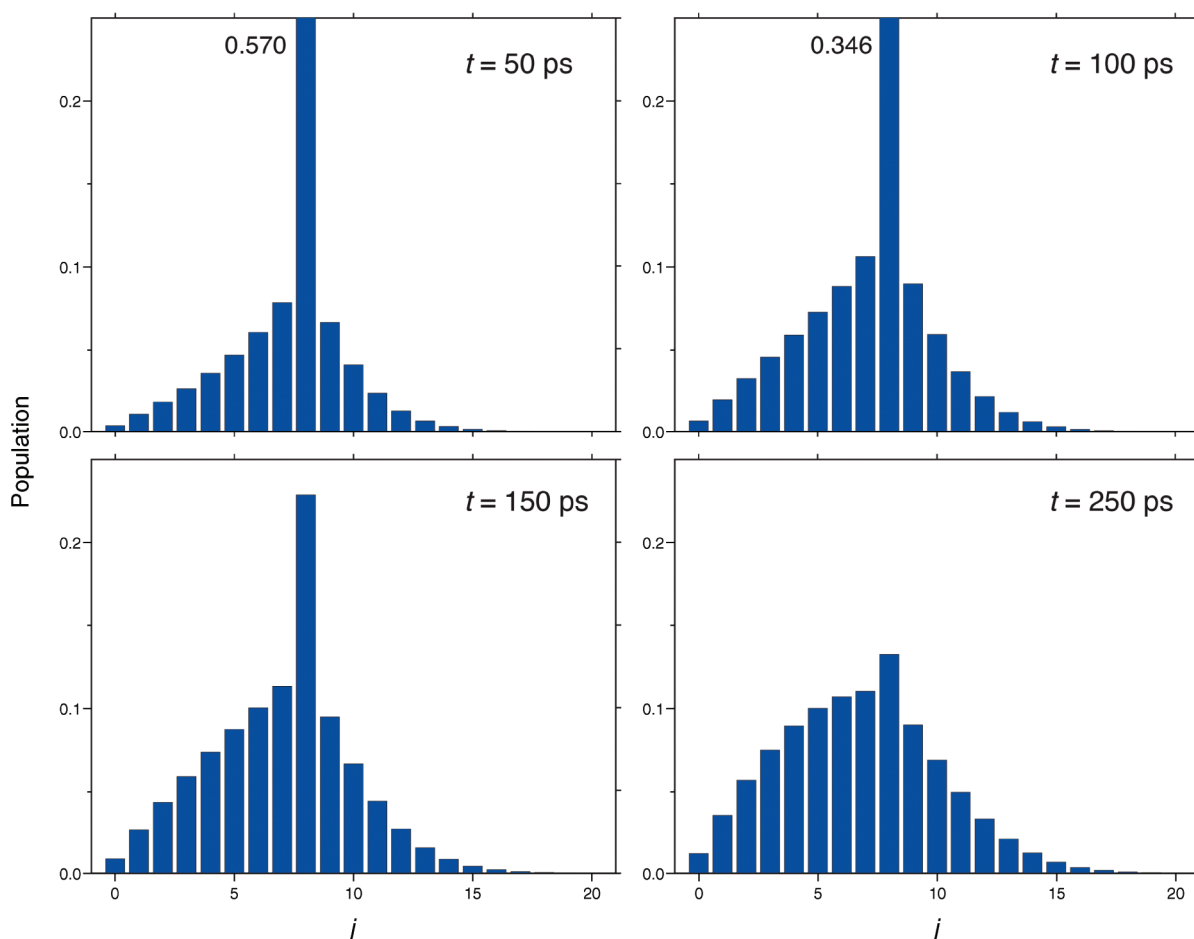


Figure 2. Results of simulation runs by the MATLAB script `master_equation_hbr.m`, using the parameters listed in Table 1. The plots display the relative number densities at 50, 100, 150, and 250 ps after initially populating just the $j = 8$ rotational level.

exponential gap approximation^{15,16} in which the lower triangle of the rate matrix is given by

$$k_{i \rightarrow j} = \alpha \exp[\beta(E_i - E_j)/k_B T], \quad j > i \quad (23)$$

where α and β are two parameters. Several workers have attempted to ascribe physical meaning to the parameters α and β , but they are best described as fitting parameters.^{17,18}

The upper triangle of the rate matrix is obtained by detailed balance (eq 5), and the diagonal elements are obtained from the conservation theorem among the elements of each column (eq 3). This model is applied in the program to a system defined by the parameters given in Table 1, which are appropriate to the relaxation of $\text{HBr}(\nu = 1)$ in helium at a total pressure of 1 atm. and a temperature of 800 K. These parameters were taken from the recent study by Kabir et al.¹⁹ While $\text{HBr}(\nu = 1)$ can also undergo collisional vibrational relaxation, the present treatment excludes this process.

The four lowest eigenvectors of the rate matrix $\mathbf{K}$ (the four lowest relaxation normal modes) are shown in Figure 1. Any initial distribution corresponds to population distributed among these (and the higher) relaxation normal modes. Each successive eigenvector has an additional sign change (node) in the underlying population distribution. At long times, only the distribution corresponding to the lowest relaxation normal mode (displayed in the top panel of Figure 1) survives. This is the Boltzmann distribution.

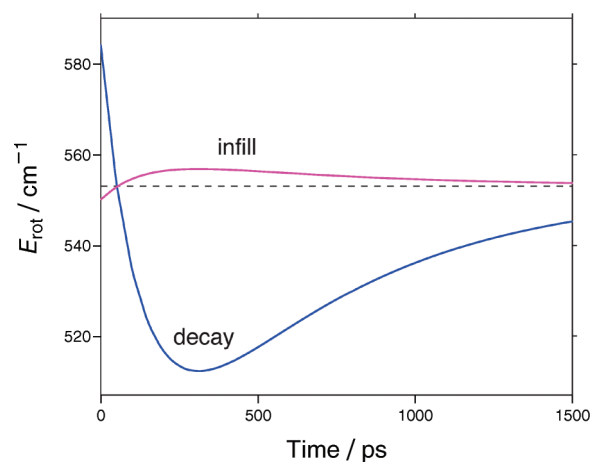


Figure 3. Relaxation of the average rotational energy after initial population of the $j = 8$ rotational level (blue curve) and after initial depletion of the $j = 8$ level from a Boltzmann distribution (magenta curve). The dashed line marks the long-time (Boltzmann) limit at $T = 800$ K.

The script determines the relative number densities in each rotational state j as a function of time, ranging from $t = 0$ to 3000 ps in steps of 50 ps for two different initial conditions: (i) when only the $j = 8$ state is populated and (ii) when all the states are

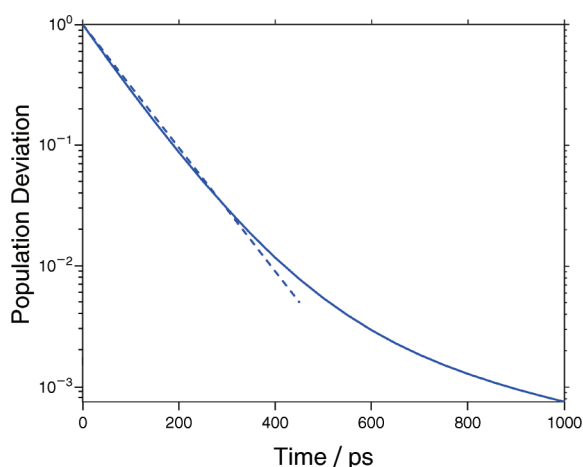


Figure 4. (Solid curve) Normalized approach to equilibrium population of the $j = 8$ rotational state for initial population of this state. An identical plot is obtained for normalized approach to equilibrium population for initial depletion of $j = 8$ out of a Boltzmann distribution. The parameters listed in Table 1 were employed in the simulation runs. The dashed line has the slope of $-\rho k(j = 8, j = 8)$, the initial decay rate for $j = 8$, neglecting reverse population transfer.

thermally populated and the $j = 8$ state is bleached. Figure 2 shows the relative populations after 50, 100, 150, and 250 ps. Figure 3 shows the time-dependent average rotational energy, which is defined as

$$\langle E_{\text{rot}} \rangle = \frac{\sum_i E_i n_i(t)}{\sum_i n_i(t)} \quad (24)$$

Strictly speaking, the denominator is not needed because the initial population distribution is assumed to be normalized. For an initial population of $j = 8$ (Figure 3), the average rotational energy initially decreases and then increases to the Boltzmann average of 556 cm^{-1} (800 K) at long times. Running the program with an initial population in a different rotational state will, of course, lead to a different time dependence of the average rotational energy.

It is also interesting to compare the time dependence of the number density of the initial state. In the case where a single rotational state is initially populated, the rate of change of the number density of this state can be approximated at short times by

$$\frac{dn_i}{dt} = -\rho k_{ii} n_i = -\rho \sum_{f \neq i} k_{i \rightarrow f} n_i \quad (25)$$

because the number densities of other states are initially small (cf eqs 1 and 4). Hence, the number density of the initially prepared state decreases with the first-order rate constant ρk_{ii} at short times. At longer times, once the number densities in the other states becomes significant, the full set of kinetic equations (eq 1) must be solved.

Consider a situation in which all the rotational states are in thermal equilibrium and a single state is depleted. The rate of change of the number density of the initially depleted state can be initially expressed as (cf eq 1)

$$\frac{dn_i}{dt} = \rho \sum_{f \neq i} k_{f \rightarrow i} n_f \quad (26)$$

because back transfer out of the depleted state can be ignored. Using detailed balance (eq 4), it can be shown that the refilling rate

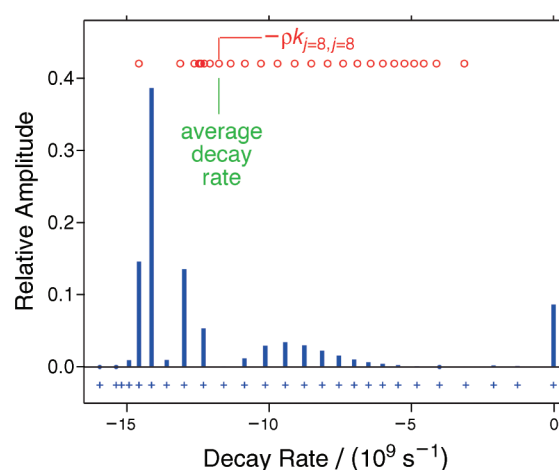


Figure 5. The multiexponential components of the $j = 8$ decay kinetics (i.e., the $j = 8$ row of eq 22, excluding the exponential) for the case of initial population of the $j = 8$ state. Also plotted is the decay rate averaged over the $j = 8$ amplitudes. Festooned across the top are the diagonal elements of the matrix $\rho \mathbf{K}$ (i.e., the negative of the overall decay rates); the diagonal $j = 8$ element of $\rho \mathbf{K}$ is indicated. The eigenrates ($\rho \mathbf{\Lambda}$) are indicated under the bars indicating their relative weights.

of the depleted state equals ρk_{ii} . Hence, the refilling rate is equal in magnitude (and opposite in sign) to the loss rate under conditions where only this single state is initially populated. Figure 3 compares the time dependence of the average rotational energy following either initial population of just the $j = 8$ level (discussed earlier) or following depletion of the $j = 8$ level from a Boltzmann distribution. In the latter case, because the distortion from the equilibrium distribution is less pronounced, the average rotational energy remains closer to its equilibrium value.

Figure 4 shows the time dependence of the deviation of the population in the $j = 8$ state from the Boltzmann (equilibrium) population compared to the initial value of this deviation, in other words

$$\delta_i(t) \equiv [n_i(t) - \Pi_{ii}(T)] / [n_i(t = 0) - \Pi_{ii}(T)] \quad (27)$$

when only the $j = 8$ state is populated initially. In the case of a depletion experiment, an identical curve describes the time-dependence of the Boltzmann (equilibrium) population compared to the initial value of this deviation. The derivative of this curve is the effective rate for the disappearance of an imposed deviation from the Boltzmann distribution. The dashed blue line in Figure 4 shows the prediction under the assumption that population loss or population infill is governed solely by the overall loss rate constant out of the state in question, k_{ii} .

The curves in Figures 3 and 4 show that the relaxation to equilibrium cannot be described completely by a single exponential. Notwithstanding, Figure 4 does reveal that roughly the first 90% of the return to equilibrium of a perturbed distribution is monoexponential, with an exponent equal to the overall loss rate constant out of the state in question. To explore this point further, the amplitudes of the multiple exponential contributions are plotted versus the total decay of population in $j = 8$ when only this state is initially populated (Figure 5). Significant contributions from many, but not all, of the eigenvalues are found, including a contribution of 0.086 from the stationary eigenvalue, an amplitude that is just the Boltzmann fraction of the state $j = 8$ for the conditions of this model. The weighted average of all the

decay rates is also indicated in Figure 5, giving a value that is not coincidentally identical to the diagonal $j = 8$ element of the original rate matrix K , which was equated above with the initial decay rate.

■ ASSOCIATED CONTENT

5 Supporting Information

Source listings of the programs; an appendix in which the authors provide a proof that the rate matrix has a nondegenerate zero eigenvalue and that its eigenvector is the Boltzmann distribution. This material is available via the Internet at <http://pubs.acs.org>.

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